SMARTBRIDGE
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SKILL WALLET
A SmartBridge Product

Coding & Solution

Date	15 March 2026
Team ID	xxxxxx
Project Name	**Human Development Index (HDI) Predictor**
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/raghva2604/Human-Development-Index-Predictor
Programming Language(s)	Python, JavaScript, HTML5, CSS3
Framework(s) Used	Flask, React.js, Vite, Tailwind CSS, Scikit-learn, Pandas, NumPy, Axios

Key Features Implemented	HDI Prediction using Machine Learning, Data Preprocessing, Input Validation, Interactive Dashboard, AI Insights, Prediction History, CSV Export, PDF Export, REST API Integration, Responsive UI
Pending / Incomplete Features	None (All planned core features implemented. Future enhancements include user authentication, real-time UNDP data synchronization, and advanced analytics.)
Setup / Run Instructions	1. Clone the repository. 2. Create a Python virtual environment. 3. Install backend requirements (pip install -r requirements.txt). 4. Run Flask backend (python app.py). 5. Install frontend dependencies (npm install). 6. Start the React application (npm run dev). 7. Access the application through the browser.

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Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows a modular architecture with separate backend, frontend, dataset, and machine learning components. The backend is implemented using Flask and Scikit-learn, while the frontend is developed using React.js and Tailwind CSS. Input validation, REST API communication, prediction history, CSV/PDF export, and interactive visualizations have been implemented to improve usability and maintainability. The project is version-controlled using